Islamic University
Faculty of Engineering
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Year No. Forth

الجامعة الإسلامية
كلية الهندسة
قسم هندسة تقنيات الحاسوب
المرحلة الرابعة

Name of lab.: Information Theory and Coding

اسم المختبر: نظرية المعلومات والترميز

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Lab. 3

2026-2027

Description of Experiment / Practice

1. **Experiment / Practice No:** 3
2. **Title of Experiment / Practice:** Mutual Information Measurements
3. **Objectives of Experiment / Practice:**
   - Understand the concept of mutual information.
   - Study the relationship between two random variables X and Y.
   - Calculate marginal probabilities from a joint probability distribution.
   - Calculate source entropy H(X), receiver entropy H(Y), and joint entropy H(X,Y).
   - Calculate mutual information I(X;Y).
   - Use MATLAB to calculate mutual information.
   - Analyze the amount of information transferred from the transmitter to the receiver.
4. **Number of Required Hours:** (2 hours)
5. **Expected Outcomes of Experiment / Practice**

   After completing this experiment, students will be able to:
   - Define mutual information.
   - Calculate marginal probabilities P(X) and P(Y).
   - Calculate H(X), H(Y), and H(X,Y).
   - Calculate mutual information between two random variables.
   - Explain the relationship between mutual information and statistical dependence.
   - Use MATLAB to measure mutual information.
   - Interpret the effect of channel uncertainty on transmitted information.
6. **Language:** English
7. **Short Explanation (theory)**

   What is Mutual Information?

   In Information Theory, Mutual Information measures the amount of information that one random variable provides about another random variable.

   For a communication system:

   - X represents the transmitted symbols.

- Y represents the received symbols.

Mutual information indicates how much knowing Y reduces the uncertainty about X.

It is denoted by:

$$I(X;Y)$$

One useful relationship is:

$$\boldsymbol{I(X;Y) = H(X) + H(Y) - H(X,Y)}$$

where:

- H(X) = Source entropy
- H(Y) = Receiver entropy
- H(X,Y) = Joint entropy
- I(X;Y) = Mutual information

The same concept is introduced in Chapter 2 through the relationships between entropy, conditional entropy, and trans-information.

Basic Idea

Mutual information measures the dependence between X and Y.

If Y gives a large amount of information about X:

$$\boldsymbol{I(X;Y)} \uparrow$$

If Y gives little information about X:

$$\boldsymbol{I(X;Y)} \downarrow$$

Therefore:

Strong relationship → Higher Mutual Information

Weak relationship → Lower Mutual Information

If X and Y are statistically independent:

$$\boldsymbol{I(X;Y) = 0}$$

For an extremely noisy channel, the received signal gives no useful information about the transmitted signal, so mutual information becomes zero.

Marginal Probabilities

Consider the following joint probability matrix:

$$\mathbf{P(X, Y)} = \begin{bmatrix} \mathbf{0.5} & \mathbf{0.1} \\ \mathbf{0.3} & \mathbf{0.1} \end{bmatrix}$$

First check:

0.5+0.1+0.3+0.1=1

Therefore, it is a valid joint probability distribution.

The marginal probability of X is obtained by summing the rows:

$$P(X) = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

because:

$P(x_1) = 0.5+0.1=0.6$

$P(x_2) = 0.3+0.1=0.4$

The marginal probability of \(Y\) is obtained by summing the columns:

P(Y) = [0.8 0.2]

because:

$P(y_1) = 0.5+0.3=0.8$

$P(y_2) = 0.1+0.1=0.2$

Source and Receiver Entropy

The source entropy is:

$$\boldsymbol{H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i)}$$

The receiver entropy is:

$$\boldsymbol{H(Y) = -\sum_{j=1}^{m} P(y_j) \log_2 P(y_j)}$$

For this example:

$$\boldsymbol{H(X) = -[0.6 \log_2 0.6 + 0.4 \log_2 0.4]}$$

$$\boldsymbol{H(X) = 0.9710\ bits/symbol}$$

and:

$$\boldsymbol{H(Y) = -[0.8 \log_2 0.8 + 0.2 \log_2 0.2]}$$

$$\boldsymbol{H(Y) = 0.7219\ bits/symbol}$$

Joint Entropy

Joint entropy is calculated using:

$$H(X,Y) = -\sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \log_2 P(x_i, y_j)$$

For the given joint probability matrix:

$$H(X,Y) = -[(0.5 \log_2(0.5)) + (0.1 \log_2(0.1)) + (0.3 \log_2(0.3)) + (0.1 \log_2(0.1))]$$
$$= 1.6855 bits/symbol$$

Mutual Information Calculation

Using:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

Substitute the calculated values:

$$I(X;Y) = 0.9710 + 0.7219 - 1.6855$$
$$= 0.0074 bits/symbol$$

This small value indicates that the received variable Y provides only a small amount of information about X.

8. Required Materials

- MATLAB Software
- Computer/Laptop
- Lecture Notes

9. Apparatus and Instruments

- Personal Computer
- MATLAB Installed Environment

10. Procedures

1. Start MATLAB.
2. Clear the command window, workspace variables, and previously opened figures.
3. Define the joint probability matrix P(X,Y).
4. Verify that the sum of all joint probabilities equals 1.
5. Calculate the marginal probability P(X).
6. Calculate the marginal probability P(Y).

7. Calculate the source entropy H(X).
8. Calculate the receiver entropy H(Y).
9. Calculate the joint entropy H(X,Y).
10. Calculate the mutual information using:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

11. Display all calculated results.
12. Interpret the mutual information value.

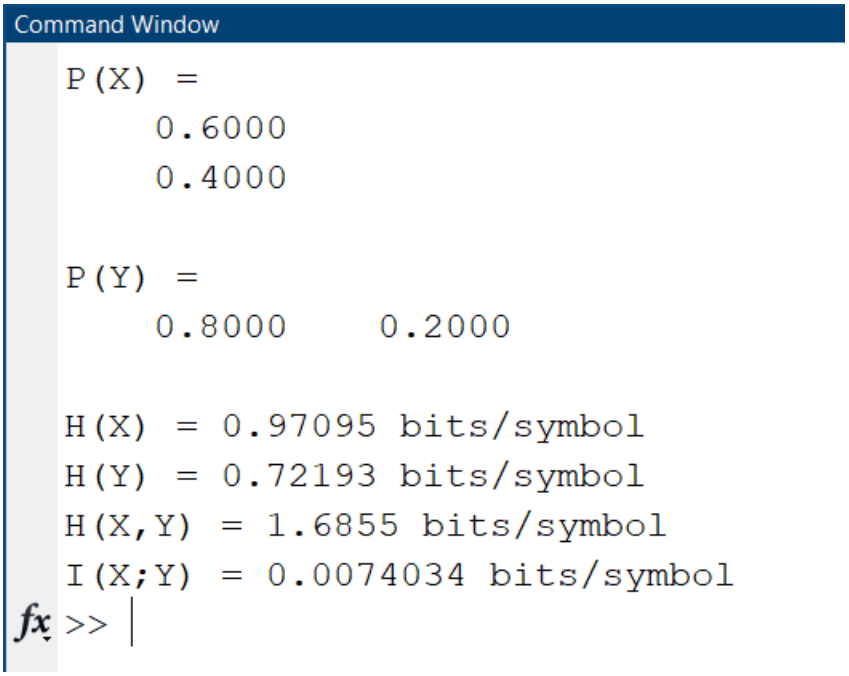
MATLAB Code:

```
clc; clear; close all;
Pxy = [0.5 0.1;
      0.3 0.1];
% Marginal probabilities
Px = sum(Pxy,2);
Py = sum(Pxy,1);
disp('P(X) = ')
disp(Px)
disp('P(Y) = ')
disp(Py)
% Entropies
Hx  = -sum(Px .* log2(Px));
Hy  = -sum(Py .* log2(Py));
Hxy = -sum(Pxy(:) .* log2(Pxy(:)));
% Mutual Information
Ixy = Hx + Hy - Hxy;
disp(['H(X) = ', num2str(Hx), ' bits/symbol'])
```

```
disp(['H(Y) = ', num2str(Hy), ' bits/symbol'])
disp(['H(X,Y) = ', num2str(Hxy), ' bits/symbol'])
disp(['I(X;Y) = ', num2str(Ixy), ' bits/symbol'])
```

11. Observation / Result

The calculated values are approximately:



The results show that the mutual information is very small.

Therefore:

Low Mutual Information → Weak Information Dependence.

12. Conclusion

In this experiment, mutual information between two random variables was calculated using MATLAB.

The marginal probabilities P(X) and P(Y) were first obtained from the joint probability matrix. Then, source entropy, receiver entropy, and joint entropy were calculated.

The mutual information was determined using:

$$\boldsymbol{I(X;Y) = H(X) + H(Y) - H(X,Y)}$$

Mutual information measures how much information the received variable \(Y\) provides about the transmitted variable X.

A larger value indicates a stronger information relationship, while:

$$\mathbf{I(X;Y) = 0}$$

indicates that the random variables are statistically independent.

13.Homework

Q1: Using MATLAB, calculate the mutual information for the following joint probability matrix:

$$P(X, Y) = \begin{bmatrix} 0.40 & 0.10 \\ 0.10 & 0.40 \end{bmatrix}$$

Find:

1. P(X)
2. P(Y)
3. H(X)
4. H(Y)
5. H(X,Y)
6. I(X;Y)

14.References

- Claude E. Shannon, A Mathematical Theory of Communication.
- Thomas M. Cover and Joy A. Thomas, Elements of Information Theory.
- MATLAB Documentation.